34.5: Amniotes are tetrapods that have a terrestrially adapted egg

The **amniotes** are a group of tetrapods whose extant members are the reptiles (including birds, as we'll discuss in this section) and mammals **(Figure 34.25)**. During their evolution, amniotes acquired a number of new adaptations to life on land.

Figure 34.25

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A phylogeny of amniotes.
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Tree diagram for the phylogeny of amniotes, with both extant and extinct groups.

Tree diagram for the phylogeny of amniotes, with both extant and extinct groups. The tree is represented by the following list.

Synapsids; Mammals; extant; example is hippopotamus

Diapsids (Branch point 2); common ancestor of reptiles

Plesiosaurs

Ichthyosaurs

Lepidosaurs (branch point 3)

Tuataras

Squamates (lizards and snakes)

Archosaurs (branch point 4)

Turtles; extant

Branch point 5

Crocodilians; extant; example is crocodile

Branch point 6

Pterosaurs

Dinosaurs (branch point 7)

Ornithischian dinosaurs

Saurischians (branch point 8)

Saurischian dinosaurs other than birds

Birds

Extant groups are named at the tips of the branches in boldface type. The dagger symbols (†) indicate extinct groups.

VISUAL SKILLS *Based on this phylogeny, are pterosaurs dinosaurs? Are birds? Explain.*

*For suggested answer, see*Appendix A.**

Derived Characters of Amniotes

Amniotes are named for the major derived character of the clade, the **amniotic egg**, which contains four specialized membranes: the amnion, the chorion, the yolk sac, and the allantois **(Figure 34.26)**. Called *extraembryonic membranes* because they are not part of the body of the embryo itself, these membranes develop from tissue layers that grow out from the embryo. The amniotic egg is named for the amnion, which encloses a compartment of fluid that bathes the embryo and acts as a hydraulic shock absorber. The other membranes in the egg function in gas exchange, the transfer of stored nutrients to the embryo, and waste storage. The amniotic egg was a key evolutionary innovation for terrestrial life: It allowed the embryo to develop on land in its own private "pond," hence reducing the dependence of tetrapods on an aqueous environment for reproduction.

Figure 34.26

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The amniotic egg.
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Diagram of embryo within shelled reptile egg and with extraembryonic membranes (amnion, yolk sac, allantois and chorion) described.

The embryos of reptiles and mammals form four extraembryonic membranes: the allantois, chorion, amnion, and yolk sac. This diagram shows these membranes in the shelled egg of a reptile.

Extraembryonic membranes are as follow.

Allantois. The allantois is a disposal sac for certain metabolic wastes produced by the embryo.

Chorion. The chorion and the membrane of the allantois exchange gases between the embryo and the air.

Amnion. The amnion protects the embryo in a fluid-filled cavity that cushions against mechanical shock.

Yolk sac. The yolk sac contains the yolk, a stockpile of nutrients. Other nutrients are stored in the albumen (egg white). Diagram of embryo within shelled reptile egg, described from the inside towards the shell. Non-membrane components: The embryo is within the amniotic cavity with amniotic fluid. The yolk provides the embryo with nutrients. Beyond the yolk is albumen; the outer layer is the shell. Extraembryonic membranes: Innermost is the yolk sac around the yolk. The amnion, chorion, and allantois are all inside the shell.

In contrast to the shell-less eggs of amphibians, the amniotic eggs of most reptiles and some mammals have a shell. A shell slows dehydration of the egg in air, an adaptation that helped amniotes to occupy a wider range of terrestrial habitats than amphibians, their closest living relatives. (Seeds played a similar role in the evolution of plants, as discussed in Concept 30.1.) Most mammals have lost the eggshell over the course of their evolution, and the embryo avoids desiccation by developing within the amnion inside the mother's body.

Amniotes have acquired other key adaptations to life on land. For example, amniotes use their rib cage to ventilate their lungs. This method is more efficient than throat-based ventilation, which amphibians use as a supplement to breathing through their skin. The increased efficiency of rib cage ventilation may have allowed amniotes to abandon breathing through their skin and develop less permeable skin, thereby conserving water.

Early Amniotes

The most recent common ancestor of living amphibians and amniotes lived 350 million years ago. Based on where their fossils have been found, the earliest amniotes lived in warm, moist areas, as did the first tetrapods. Over time, early amniotes expanded into a wide range of new environments, including dry and high-latitude regions. Fossil evidence shows that the earliest amniotes resembled small lizards with sharp teeth, a sign that they were predators **(Figure 34.27)**. Later groups of amniotes also included herbivores, as evidenced by their grinding teeth and other features.

Figure 34.27

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Artist's reconstruction of *Hylonomus*, an early amniote.
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Hylonomus resembles a lizard.

About 25 cm long, this species lived 310 million years ago and probably ate insects and other small invertebrates.

Present-day amniotes include two large clades of terrestrial vertebrates: reptiles and mammals.

Reptiles

An evolutionary tree highlights reptilia.

Branch point 1: First level-1 branch: Cephalochordata. The second level-1 branch splits from branch point 2 into the first level-2 branch: Urochordata. The second level-2 branch splits from branch point 3 into the first and second level-3 branches. The first level-3 branch splits from branch point 4 into the first level-4 branch: Myxini and the second level-4 branch: Petromyzontida. The second level-3 branch splits from branch point 5 into the first level-5 branch: Chondrichthyes. The second level-5 branch splits from branch point 6 into the first level-6 branch: Actinopterygii. The second level-6 branch splits from branch point 7 into the first level-7 branch: Actinista. The second level-7 branch splits from branch point 8 into the first level-8 branch: Dipnoi. The second level-8 branch splits from branch point 9 into the first level-9 branch: Amphibia. The second level-9 branch splits from branch point 10 into the first level-10 branch: Reptilia and the second level-10 branch: Mammalia.

Living members of the **reptile** clade include turtles, tuataras, lizards and snakes, crocodilians, and birds (see Figure 34.25). There are about 20,800 species of reptiles, the majority of which are squamates (lizards and snakes; 10,425 species) or birds (10,000 species).

As a group, the reptiles share several derived characters that distinguish them from other tetrapods. For example, unlike amphibians, reptiles have scales that contain the protein keratin (as does a human nail). Scales help protect the animal's skin from desiccation and abrasion. In addition, most reptiles lay their shelled eggs on land; the shell protects the egg from drying out **(Figure 34.28)**. Fertilization occurs internally, before the eggshell is secreted.

Figure 34.28

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Hatching reptiles.
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Baby panther chameleons break out of their parchment-like shells.

These baby panther chameleons (Furcifer pardalis) are breaking out of their parchment-like shells, a common type of shell among living reptiles other than birds.

Reptiles such as lizards and snakes are sometimes described as "cold-blooded" because they do not use their metabolism extensively to control their body temperature. However, they do regulate their body temperature by using behavioral adaptations. For example, many lizards bask in the sun when the air is cool and seek

shade when the air is too warm. A more accurate description of these reptiles is to say that they are **ectothermic**, which means that they absorb external heat as their main source of body heat. By warming themselves directly with solar energy rather than through the metabolic breakdown of food, an ectothermic reptile can survive on less than 10% of the food energy required by a mammal of the same size. But the reptile clade is not entirely ectothermic; birds are **endothermic**, capable of maintaining body temperature through metabolic activity.

The Origin and Evolutionary Radiation of Reptiles

Fossil evidence indicates that the earliest reptiles lived about 310 million years ago and resembled lizards. Like all reptiles today, these early reptiles were **diapsids**. A key derived character of diapsids is a pair of holes on each side of the skull, behind the eye sockets; muscles pass through these holes and attach to the jaw, controlling jaw movement.

The diapsids are composed of three main lineages. The first lineage includes the turtles. A second lineage gave rise to the **lepidosaurs**, which include tuataras, lizards, and snakes. This lineage also produced some marine reptiles, including the giant mososaurs. Some of these marine species rivaled today's whales in length; all of them are extinct. The third diapsid lineage, the **archosaurs**, produced the crocodilians, pterosaurs, and dinosaurs. Our focus here will be on extinct lineages of archosaurs; we'll discuss living reptiles shortly.

Pterosaurs, which originated in the late Triassic, were the first tetrapods to exhibit flapping flight. The pterosaur wing was completely different from the wings of birds and bats. It consisted of a collagen-strengthened membrane that stretched between the trunk or hind leg and a very long digit on the foreleg. The smallest pterosaurs were no bigger than a sparrow, and the largest had a wingspan of nearly 11 m. They appear to have converged on many of the ecological roles later played by birds; some were insect-eaters, others grabbed fish out of the ocean, and still others filtered small animals through thousands of fine needlelike teeth. But by 66 million years ago, pterosaurs had become extinct.

On land, the **dinosaurs** diversified into a vast range of shapes and sizes, from bipeds the size of a pigeon to 45-m-long quadrupeds with necks long enough to let them browse the tops of trees. One lineage of dinosaurs, the ornithischians, were herbivores; they included many species with elaborate defenses against predators, such as tail clubs and horned crests. The other main lineage of dinosaurs, the saurischians, included the long-necked giants and a group called the **theropods**, which were bipedal carnivores. Theropods included the famous *Tyrannosaurus rex* as well as the ancestors of birds.

Dinosaurs once were considered slow, sluggish creatures. Since the 1970s, however, fossil discoveries and research have led to the conclusion that many dinosaurs were agile and fast moving. Dinosaurs had a limb structure that enabled them to walk and run more efficiently than could earlier tetrapods, which had a

sprawling gait. Fossilized footprints and other evidence suggest that some species were social—they lived and traveled in groups, much as many mammals do today. Paleontologists have also discovered evidence that some dinosaurs built nests and protected their eggs, as birds do today (see Figure 26.17). Finally, some anatomical evidence supports the hypothesis that at least some dinosaurs were endotherms.

All dinosaurs except birds became extinct by the end of the Cretaceous period (66 million years ago). Their extinction may have been caused at least in part by the asteroid or comet impact described in Concept 25.4. Some analyses of the fossil record are consistent with this idea in that they show a sudden decline in dinosaur diversity at the end of the Cretaceous. However, other analyses indicate that the number of dinosaur species had begun to decline several million years before the Cretaceous ended. Further fossil discoveries and new analyses will be needed to resolve this debate.

Next, we'll discuss the three extant lineages of reptiles, the turtles, lepidosaurs (tuataras, lizards, and snakes), and archosaurs (crocodilians and birds).

Interview

Interview with Paul Sereno: Adventures hunting dinosaurs

A photograph of Paul Sereno, an American paleontologist.

HHMI Video

How to Find a Dinosaur

image

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Turtles

Turtles are one of the most distinctive groups of reptiles alive today. For example, turtles do not have any holes in their skull behind the eye sockets, whereas other reptiles have two holes behind each eye socket. Recall that such skull holes are a key derived trait of the diapsids. Thus, until recently it was not clear whether turtles—like all other living reptiles—should be classified within the diapsid clade. However, in 2015, new fossil discoveries showed that early turtles had the skull openings found in other diapsids. This suggests that turtles are diapsids that have lost the holes in their skull over the course of evolution. Moreover, recent phylogenomic studies indicate that turtles are nested within the diapsid clade and are more closely related to crocodilians and birds than to other reptiles (see Figure 34.25).

All turtles have a boxlike shell made of upper and lower shields that are fused to the vertebrae, clavicles (collarbones), and ribs **(Figure 34.29a)**. Most of the 351 known species of turtles have a hard shell, providing excellent defense against predators. Fossil evidence shows that *Pappochelys*, a turtle that lived 240 million years ago, had a series of hard, shell-like bones along its belly. By 220 million years ago, another early turtle had a complete lower shell but an incomplete upper shell, suggesting that turtles acquired full shells in stages.

Figure 34.29

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Extant reptiles (other than birds).
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Black-breasted hill turtle moves through grass. Tuatara resembling an iguana on a rock. Wagler's pit viper coils around a tree limb. Australian thorny devil lizard extends legs to lift spiny body off sand. American alligator climbs out of the water.

The earliest turtles could not retract their head into their shell, but mechanisms for doing so evolved independently in two separate branches of turtles. The side-necked turtles fold their neck horizontally, while the vertical-necked turtles fold their neck vertically.

Some turtles have adapted to deserts, and others live almost entirely in ponds and rivers. Still others live in the sea. Sea turtles have a reduced shell and enlarged forelimbs that function as flippers. They include the largest living turtles, the deep-diving leatherbacks, which can exceed a mass of 1,500 kg and feed on jellies. Leatherbacks and other sea turtles are endangered by being caught in fishing nets, as well as by the residential and commercial development of the beaches where the turtles lay their eggs.

Lepidosaurs

One surviving lineage of lepidosaurs is represented by a single species, a lizard-like reptile called the tuatara **(Figure 34.29b)**. Fossil evidence indicates that tuatara ancestors lived at least 220 million years ago. These organisms thrived on many continents well into the Cretaceous period and reached up to a meter in length. Today, however, tuataras are found only on 30 islands off the coast of New Zealand. When humans arrived in New Zealand 750 years ago, the rats that accompanied them devoured tuatara eggs, eventually eliminating the reptiles on the main islands. The tuataras that remain on the outlying islands are about 50 cm long and feed on insects, small lizards, and bird eggs and chicks. They can live to be over 100 years old. Their future survival depends on whether their remaining habitats are kept rat-free.

The other major living lineage of lepidosaurs consists of the lizards and snakes, or squamates, which number 10,425 species **(Figure 34.29c and d)**. Many squamates are small; the Jaragua lizard, discovered recently

in the Dominican Republic, is only 16 mm long—small enough to fit comfortably on a dime. In contrast, the Komodo dragon of Indonesia is a lizard that can reach a length of 3 m. It hunts deer and other large prey, delivering venom with its bite.

Snakes descended from lizards with legs (see Figure 26.1); thus they are classified as legless lizards. Today, some species of snakes retain vestigial pelvic and limb bones, providing evidence of their ancestry. Despite their lack of legs, snakes are quite proficient at moving on land, most often by producing waves of lateral bending that pass from head to tail. Force exerted by the bends against solid objects pushes the snake forward. Snakes can also move by gripping the ground with their belly scales at several points along the body while the scales at intervening points are lifted slightly off the ground and pulled forward.

Snakes are carnivorous, and a number of adaptations aid them in hunting and eating prey. They have acute chemical sensors, and though they lack eardrums, they are sensitive to ground vibrations, which helps them detect the movements of prey. Heat-detecting organs between the eyes and nostrils of pit vipers, including rattlesnakes, are sensitive to minute temperature changes, enabling these night hunters to locate warm animals. Venomous snakes inject their toxin through a pair of sharp teeth that may be hollow or grooved. The flicking tongue is not venomous but helps fan odors toward olfactory (smell) organs on the roof of the mouth. Loosely articulated jawbones and elastic skin enable most snakes to swallow prey larger than the diameter of the snake's head (see Figure 23.14).

Next, we'll turn to the third (and final) clade of reptiles with living members, the archosaurs. There are two extant groups of archosaurs: the crocodilians and the birds.

Crocodilians

Alligators and crocodiles (collectively called crocodilians) belong to a lineage that reaches back to the late Triassic. The earliest members of this lineage were small terrestrial quadrupeds with long, slender legs. Later species became larger and adapted to aquatic habitats, breathing air through their upturned nostrils. Some Mesozoic crocodilians grew as long as 12 m and may have attacked dinosaurs and other prey at the water's edge.

The 24 known species of living crocodilians are confined to warm regions of the globe. In the southeastern United States, the American alligator **(Figure 34.29e)** has made a comeback after spending years on the endangered species list.

Birds

There are about 10,000 species of birds in the world. Like crocodilians, birds are archosaurs, but almost every feature of their anatomy has been modified in their adaptation to flight **(Figure 34.30)**.

Figure 34.30

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Form fits function: the avian wing and feather.
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The avian wing and feather.

(a) *A wing is a remodeled version of the tetrapod forelimb.* ***(b)*** *The bones of many birds have a honeycombed internal structure and are filled with air.* ***(c)*** *A feather consists of a central air-filled shaft, from which radiate the vanes. The vanes are made up of barbs, which bear small branches called barbules. Birds have contour feathers and downy feathers. Contour feathers are stiff and contribute to the aerodynamic shapes of the wings and body. Their barbules have hooks that cling to barbules on neighboring barbs. When a bird preens, it runs the length of each contour feather through its beak, engaging the hooks and uniting the barbs into a precise shape. Downy feathers lack hooks, and the free-form arrangement of their barbs produces a fluffiness that provides insulation by trapping air.*

Parts are as follows.

Part a. A wing is a remodeled version of the tetrapod forelimb. Diagram shows a bird wing with bone structure detailed. In the wing, a large bone connects to the forearm, consisting of two parallel bones. These bones connect to the wrist joint, a structure composed of several small bones. Connected to the wrist, towards the tip of the wing, is a short bone (Finger 1); and the palm, consisting of two longer parallel bones. Connected to the palm towards the wing tip is Finger 2 and a shorter Finger 3.

Part b. Photograph of interior of wing bone reveals many spaces within the bone structure. The bones of many birds have a honeycombed internal structure and are filled with air.

Part c. Wing feather structure diagram consists of a thicker hollow central shaft, from which branch barbs (perpendicular to the shaft). From the barbs branch barbules (perpendicular to the barbs) with hooks.

A feather consists of a central air-filled shaft, from which radiate the vanes. The vanes are made up of barbs, which bear small branches called barbules. Birds have contour feathers and downy feathers. Contour feathers are stiff and contribute to the aerodynamic shapes of the wings and body. Their barbules have hooks that cling to the barbules on neighboring barbs. When a bird preens, it runs the length of each contour feather through its beak, engaging the hooks and uniting the barbs into a precise shape. Downy feathers lack hooks, and the free-form arrangement of their barbs produces a fluffiness that provides insulation by trapping air.

Video

Soaring Hawk

image

Derived Characters of Birds

Many of the characters of birds are adaptations that facilitate flight, including weight-saving modifications that make flying more efficient. For example, birds lack a urinary bladder, and the females of most species have only one ovary. The gonads of both females and males are usually small, except during the breeding season, when they increase in size. Living birds are also toothless, an adaptation that trims the weight of the head.

A bird's most obvious adaptations for flight are its wings and feathers (see Figure 34.30). Feathers are made of the protein β-keratin, which is also found in the scales of other reptiles. The shape and arrangement of the feathers form the wings into airfoils, and they illustrate some of the same principles of aerodynamics as the wings of an airplane. Power for flapping the wings comes from contractions of large pectoral (breast) muscles anchored to a keel on the sternum (breastbone). Some birds, such as eagles and hawks, have wings adapted for soaring on air currents and flap their wings only occasionally; other birds, including hummingbirds, must flap continuously to stay aloft (see Figure 34.34). Among the fastest birds are the appropriately named swifts, which can fly at speeds up to 170 km/hr.

Flight provides numerous benefits. It enhances scavenging and hunting, including enabling many birds to feed on flying insects, an abundant, nutritious food resource. Flight also provides ready escape from earthbound predators and enables some birds to migrate great distances to exploit different food resources and seasonal breeding areas.

Flying requires a great expenditure of energy from an active metabolism. Birds are endothermic; they use their own metabolic heat to maintain a high, constant body temperature. Feathers and in some species a layer of fat provide insulation that enables birds to retain body heat. The lungs have tiny tubes leading to and from elastic air sacs that improve airflow and oxygen uptake. This efficient respiratory system and a circulatory system with a four-chambered heart keep tissues well supplied with oxygen and nutrients, supporting a high rate of metabolism.

Flight also requires both acute vision and fine muscle control. Birds have color vision and excellent eyesight. The visual and motor areas of the brain are well developed, and the brain is proportionately larger than those of amphibians and nonbird reptiles.

Birds generally display very complex behaviors, particularly during breeding season, when they engage in elaborate courtship rituals. Because eggs have shells by the time they are laid, fertilization must be internal. Copulation usually involves contact between the openings to the birds' cloacas. After eggs are laid, the avian embryo must be kept warm through brooding by the mother, the father, or both, depending on the species.

The Origin of Birds

Cladistic analyses of birds and reptilian fossils indicate that birds belong to the group of bipedal saurischian dinosaurs called theropods. Since the late 1990s, Chinese paleontologists have unearthed a spectacular trove of feathered theropod fossils that are shedding light on the origin of birds. Several species of dinosaurs closely related to birds had feathers with vanes, and a wider range of species had filamentous feathers. Such findings imply that feathers evolved long before powered flight. Among the possible functions of these early feathers were insulation, camouflage, and courtship display.

By about 160 million years ago, a lineage of feathered theropods had evolved into birds. Many researchers consider *Archaeopteryx*, which was discovered in a German limestone quarry in 1861, to be the earliest known bird **(Figure 34.31)**. It had feathered wings but retained ancestral characters such as teeth, clawed digits in its wings, and a long tail. *Archaeopteryx* flew well at high speeds, but unlike a present-day bird, it could not take off from a standing position. Fossils of later birds from the Cretaceous show a gradual loss of certain ancestral dinosaur features, such as teeth and clawed forelimbs, as well as the acquisition of innovations found in extant birds, including a short tail covered by a fan of feathers.

Figure 34.31

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Artist’s reconstruction of the early bird *Archaeopteryx*.
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Drawing of Archaeopteryx, which is a transitional organism between reptiles and birds.

Fossil evidence indicates that Archaeopteryx was capable of powered flight but retained many characters of nonbird dinosaurs. Colors in this drawing were based on pigments extracted from fossilized Archaeopteryx feathers.

The image resembles a bird with airfoil wings in turn with contour feathers, wing claws toothed beak and a long tail with many vertebrae.

HHMI BioInteractive Video

Great Transitions: The Origin of Birds

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Living Birds

Clear evidence of Neornithes, the clade that includes the 28 orders of living birds, can be found before the Cretaceous-Paleogene boundary 66 million years ago. Several groups of living and extinct birds include one or more flightless species. The **ratites**, an order of birds that includes the ostrich, rhea, kiwi, cassowary, and

emu, are all flightless **(Figure 34.32)**. In ratites, the sternal keel is absent, and the pectoral muscles are small relative to those of birds that can fly.

Figure 34.32

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    An emu (*Dromaius novaehollandiae*), a flightless bird
native to Australia.
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Photograph of an emu walking in a field. Emu has a large feathered body, long neck, and long, thick legs.

Penguins make up another flightless order of birds, but, like flying birds, they have powerful pectoral muscles. They use these muscles to “fly” in the water: As they swim, they flap their flipper-like wings in a manner that resembles the flight stroke of a more typical bird **(Figure 34.33)**. Certain species of rails, ducks, and pigeons are also flightless.

Figure 34.33

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    A king penguin (*Aptenodytes patagonicus*) “flying”
underwater.
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Underwater photograph of a king penguin swimming with both wings outstretched.

With their streamlined shape and powerful pectoral muscles, penguins are fast and agile swimmers.

Although the demands of flight have rendered the general body forms of many flying birds similar to one another, experienced bird-watchers can distinguish species by their profile, colors, flying style, behavior, and beak shape. The skeleton of a hummingbird’s wing is unique, making it the only bird that can hover and fly backward **(Figure 34.34)**. Adult birds lack teeth, but during the course of avian evolution their beaks have taken on a variety of shapes suited to different diets. Some birds, such as parrots, have crushing beaks with which they can crack open hard nuts and seeds. Other birds, such as flamingoes, are filter feeders. Their beaks have “strainers” that enable them to capture food particles from the water **(Figure 34.35)**. Foot structure, too, shows considerable variation. Various birds use their feet for perching on branches **(Figure 34.36)**, grasping food, defense, swimming or walking, and even courtship (see Figure 24.3e).

Figure 34.34

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Hummingbird feeding while hovering.
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Photograph of a hummingbird hovering while feeding from a flower.

A hummingbird can rotate its wings in all directions, enabling it to hover and fly backward.

Figure 34.35

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A specialized beak.
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Flamingo has long legs and a thick, curved beak for scooping water.

This greater flamingo (Phoenicopterus roseus) dips its beak into the water and strains out the food.

Figure 34.36

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Feet adapted to perching.
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Photograph of the great tit, a small bird with feet wrapped tightly around a section of the barbed wire on a fence.

This great tit (Parus major) is a member of the Passeriformes, the perching birds. The toes of these birds can lock around a branch or wire, enabling the bird to rest for long periods.